

# Design, Simulation, and Experimental Verification of Mach-Zehnder Interferometers on SOI

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**Abstract**—This report presents the design, simulation, and experimental verification of nine Mach-Zehnder interferometers (MZIs) fabricated on a silicon-on-insulator (SOI) platform. The devices include variations in polarization, waveguide width, and path length difference, enabling evaluation of substrate variability and geometric sensitivity. Simulation results for grating couplers, transfer functions, and free spectral range (FSR) are compared with experimental measurements of a representative MZI, demonstrating good agreement. Group-index corner analysis is used to validate both theoretical and measured results.

**Index Terms**—Mach-Zehnder interferometer, silicon photonics, SOI, grating coupler, FSR, waveguide.

## I. INTRODUCTION

Silicon photonics enables dense integration of passive and active optical components using CMOS-compatible fabrication. Among its fundamental building blocks, the Mach-Zehnder interferometer (MZI) is used extensively in filters, switches, phase shifters, and modulators.

In this project, nine different MZI designs were implemented on a 220 nm SOI platform with 500 nm, 450 nm, and 350 nm wide strip waveguides. The designs include TE and TM operation, multiple path length differences, and waveguide-width variations. A final broadband MZI includes a directional coupler and multiple output ports.

## II. LAYOUT OVERVIEW

All nine designs were placed within the allocated chip space, including systematic placement of pairs of identical MZIs in opposite corners to quantify substrate and etch-related variability. The full layout, designed in KLayout, is shown in Fig. 1.

## III. THEORETICAL BACKGROUND

### A. MZI Operating Principles

A standard unbalanced MZI consists of two 50:50 splitters connected by arms of different length. The intensity at the output is:

$$I_o = \frac{1}{2} I_i [1 + \cos(\beta \Delta L)], \quad (1)$$

where  $\beta = 2\pi n_{\text{eff}}/\lambda$ , and  $n_{\text{eff}}$  is the mode effective index.

The free spectral range is:

$$FSR = \frac{\lambda^2}{n_g \Delta L}, \quad (2)$$

with  $n_g$  the group index.

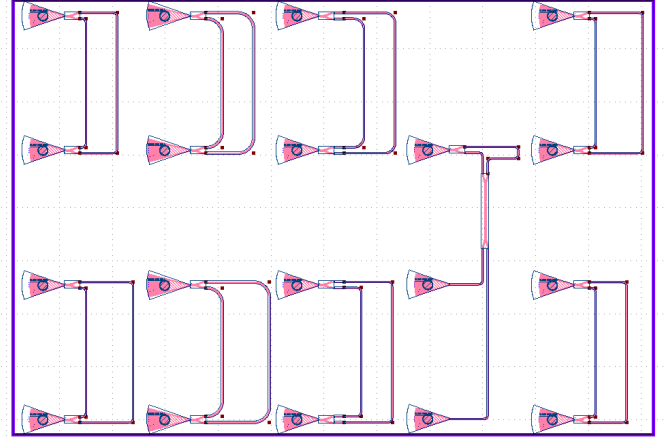


Fig. 1. KLayout view of the fabricated MZI structures, including TE/TM variants, width variations, and broadband MZI.

TABLE I  
SUMMARY OF FABRICATED MZI DESIGNS

| ID | Pol. | $\Delta L$ ( $\mu\text{m}$ ) | W (nm) | $R_b$ ( $\mu\text{m}$ ) | Notes             |
|----|------|------------------------------|--------|-------------------------|-------------------|
| 1  | TE   | 70                           | 500    | 5                       | Top-left          |
| 2  | TE   | 70                           | 500    | 5                       | Bot-right         |
| 3  | TE   | 100                          | 500    | 5                       | Bot-left          |
| 4  | TE   | 100                          | 500    | 5                       | Top-right         |
| 5  | TM   | 70                           | 500    | 15                      | Pol. test         |
| 6  | TM   | 100                          | 500    | 15                      | Pol. test         |
| 7  | TE   | 70                           | 450    | 10                      | Width var.        |
| 8  | TE   | 70                           | 350    | 5                       | Width var.        |
| 9  | TE   | —                            | 500    | —                       | Broadband coupler |

### B. Waveguide Geometry and Bend Radii

All waveguides are 220 nm tall SOI strip waveguides, using widths of 500, 450, and 350 nm. Bend radii were chosen to minimize radiation loss:

- TE, 500 nm width:  $5 \mu\text{m}$  - TM, 500 nm width:  $15 \mu\text{m}$  -  
TE, 450 nm width:  $10 \mu\text{m}$  - TE, 350 nm width:  $5 \mu\text{m}$

### C. Corner Analysis

Width variations of  $\pm 20$  nm and thickness variations of 203–212 nm were used to compute group-index extreme corners. These serve as theoretical bounds for later experimental comparison.



Fig. 2. Simulated TE grating-coupler insertion loss.

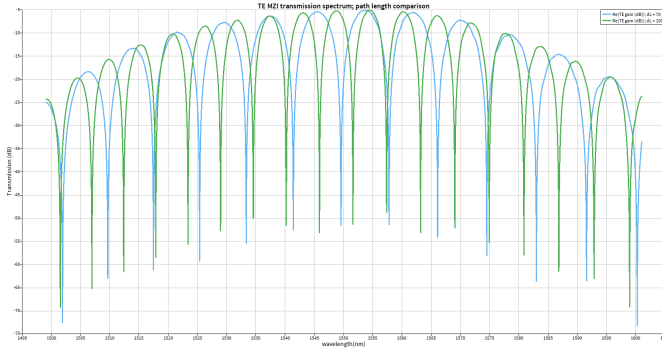


Fig. 3. Simulated MZI transfer functions for 70  $\mu\text{m}$  and 100  $\mu\text{m}$  path differences.

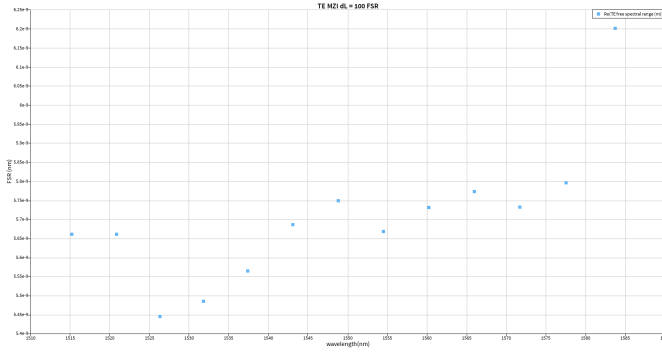


Fig. 4. Simulated FSR for the 100  $\mu\text{m}$  TE MZI.

## IV. SIMULATION RESULTS

### A. Grating Couplers

Figure 2 shows the simulated TE grating-coupler insertion loss.

### B. MZI Transfer Functions

Transfer functions for  $\Delta L = 70 \mu\text{m}$  and  $100 \mu\text{m}$  MZIs are shown in Fig. 3.

### C. FSR Simulation

The simulated FSR for the 100  $\mu\text{m}$  device is shown in Fig. 4.

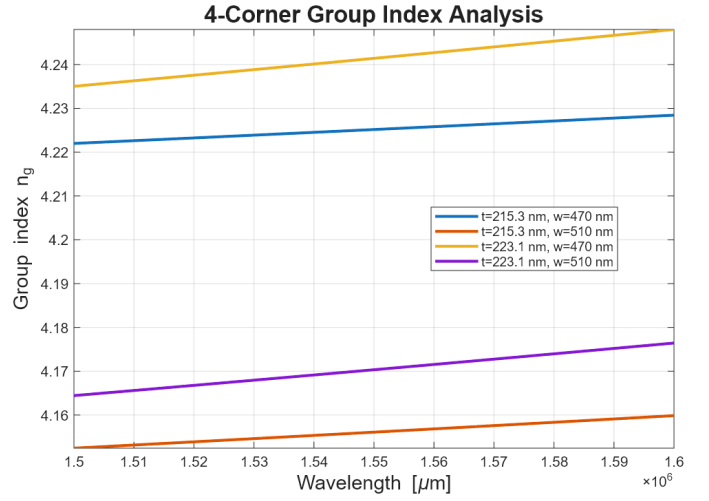


Fig. 5. Simulated group-index corner analysis for TE and TM waveguides.

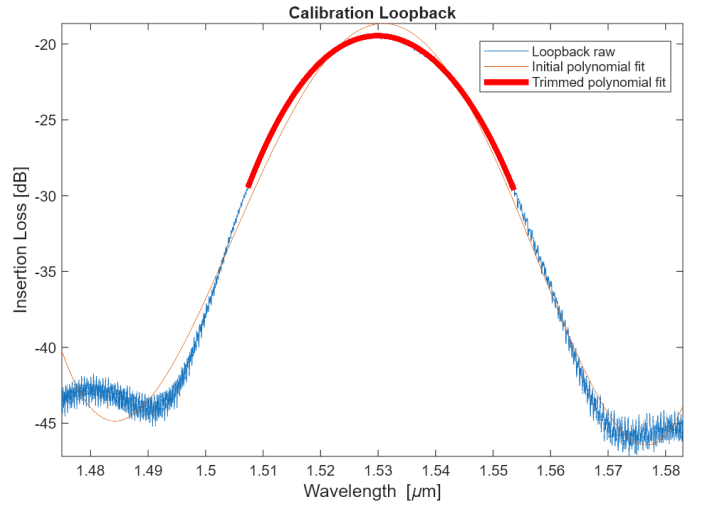


Fig. 6. Loopback structure used for de-embedding grating-coupler loss.

## D. Group Index Corner Analysis

## V. EXPERIMENTAL RESULTS

### A. Test Setup and Loopback De-embedding

A loopback waveguide (Fig. 6) was included to isolate grating-coupler loss and extract the MZI-only spectrum.

### B. Measured MZI Spectra

Raw transmission of the TE, 100  $\mu\text{m}$  device is shown in Fig. 7. The de-embedded spectrum is shown in Fig. 8.

### C. Model Fit and FSR Extraction

Figure 9 compares measured transmission with a fitted model and shows a good match. The Figure 10 shows a calculated FSR of the same MZI ( $dL=100$  microns) and visually comparing it to the simulated values we can conclude that they match.

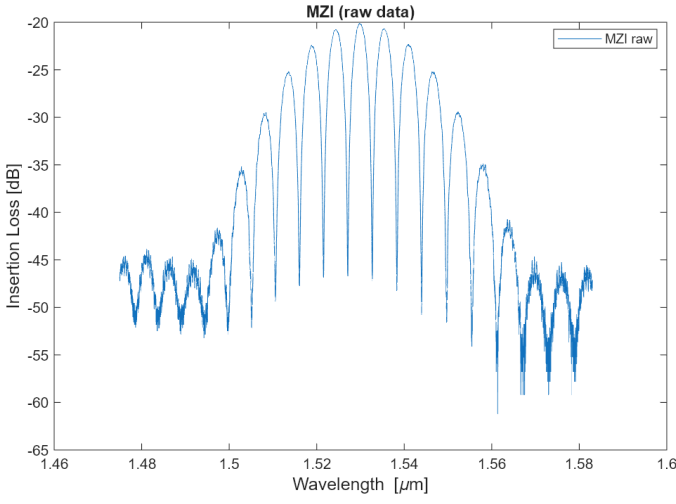


Fig. 7. Raw measured spectrum of TE MZI ( $\Delta L = 100 \mu\text{m}$ ).

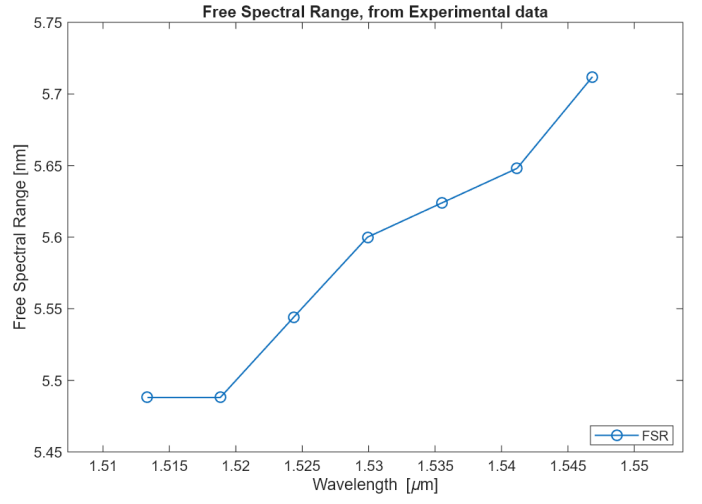


Fig. 10. Experimentally estimated FSR for the TE  $100 \mu\text{m}$  MZI.

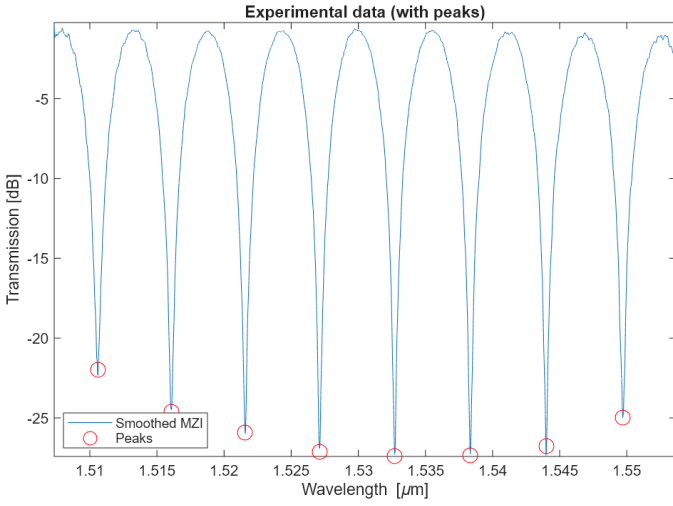


Fig. 8. De-embedded transmission spectrum after loopback correction.

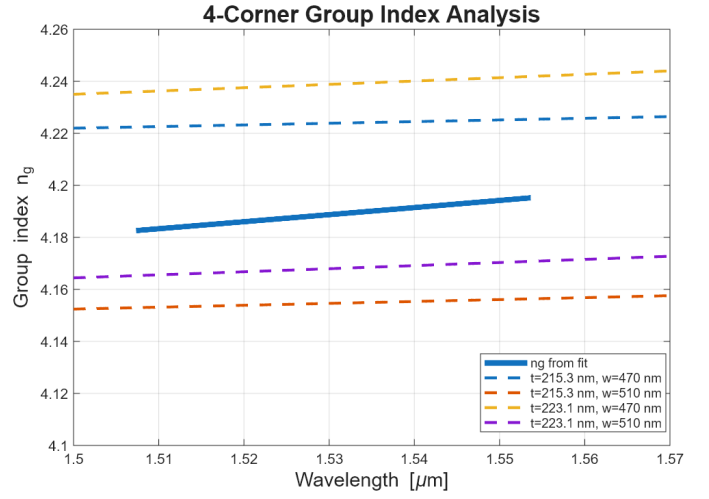


Fig. 11. Comparison of experimental and simulated group index.

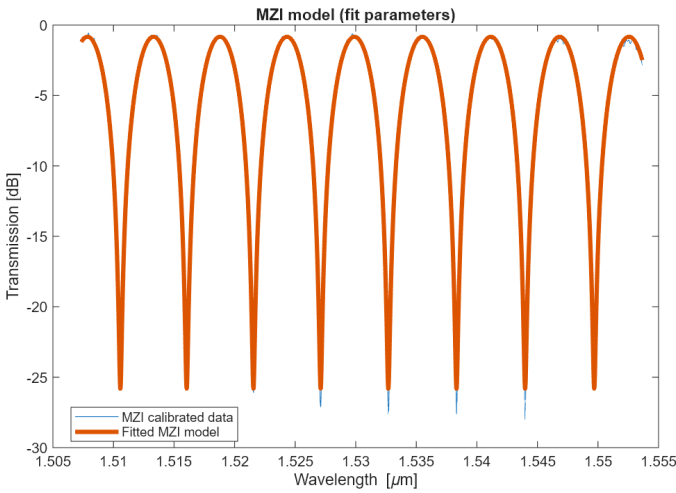


Fig. 9. Measured and fitted transmission spectrum of TE MZI ( $\Delta L = 100 \mu\text{m}$ ).

#### D. Group Index Comparison

Figure 11 shows the experimentally obtained group index in comparison with the four corner analysis. It is evident that the measured group index falls within the four corner bounds.

### VI. DISCUSSION

Simulated and experimental spectra show close agreement in fringe spacing and modulation depth. The experimental FSR corresponds with the simulated one and the experimental group index lies within the predicted corner-analysis bounds, confirming that the fabricated waveguides match design expectations within normal process variations.

### VII. CONCLUSION

A full set of nine MZIs were designed, simulated, and partially verified experimentally. The results validate the modeling methodology and the SOI platform's consistency. Future work includes measurement of all remaining devices and statistical analysis of corner-to-corner variation.

## REFERENCES

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